Two-Layer Intracellular Protein Surveillance

Conceptual architecture, mathematical hypotheses, limitations, and research roadmap — Research note v0.3 | 7 November 2026

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Scientific status: An exploratory computational hypothesis, not an experimentally validated biological system. This document does not provide clinical advice or a molecular intervention protocol.

Abstract

The proposal compares two hypothetical observation channels supplementing natural cellular quality control. Observer A generates early alerts from mRNA/translation-associated signals; observer B independently samples produced proteins and cellular context. The goal is to evaluate detection benefit against false alarms, sampling costs, and intervention risks. Four simulation arms are compared: natural surveillance (N), N+A, N+B, and N+A+B.

Origin and evolution of the concept

An initial synthetic-ribosome pre-check idea was reframed as an early-warning observer, since protein function and danger cannot reliably be inferred from sequence alone. A second observer B was introduced to inspect later protein-level evidence. Passive Brownian access, observer multiplicity, transient signaling, and the potential harms of suppressing harmless variation are explicit design considerations.

Observer A — early warning

A is a hypothetical translation-associated risk observer, not a demonstrated synthetic ribosome. Illustrative, uncalibrated score thresholds are 0.65 for natural QC alert, 0.75 for temporary translation-slowdown request, and 0.90 for a strong alert requiring B corroboration. These scores are not calibrated disease probabilities.

Observer B — independent sampling

B samples downstream protein/context information and may operate without A. Illustrative thresholds are 0.60 for increased surveillance, 0.75 for verification of a reversible response, and 0.95 for consideration of a joint candidate decision. The present computational approximation uses independent sampling opportunities rather than explicit three-dimensional Brownian trajectories.

Mathematical core

Hidden hazard state $H \in \{0,1\}$ is distinguished from benign variation and physiological cell death. Observer scores r_A and r_B are synthetic, uncalibrated risk scores. A transient signal follows $dS/dt = Q_A(t) - k_d S$, where $k_d = \ln(2)/t_{half}$. Its characteristic length is $\ell = \sqrt{D/k_d}$. A provisional geometric criterion is $\ell \geq 0.2\ L$, with L the characteristic linear cell dimension; this is not a 20% cytoplasmic volume or guaranteed delivery. For N_B independent equal sampling opportunities, P(at

least one sample) = 1 − (1−p)^N_B. A joint candidate flag is C = 1[r_A≥0.90] × 1[r_B≥0.95] × 1[message detected], with additional coverage/sampling constraints in the computational design. C is not an actual apoptosis command.

Signaling and decision safeguards

A short-lived A-to-B alert and a separately tracked B alarm are proposed. Hysteresis is represented by S_on > S_off, but a full dynamic switching state machine has not yet been implemented. A alone cannot trigger an irreversible candidate decision when B disagrees. A high joint score remains only a simulation label, not a biological actuator.

Comparative computational evaluation

Evaluate N, N+A, N+B, and N+A+B on matched virtual-cell samples. Compare sensitivity, false-positive rate, missed hazards, candidate-decision frequency, sampling burden, and cost under matched resource budgets. Inputs and results are synthetic rather than measurements. Correlated errors between observers may erase the expected benefit of corroboration.

Limitations and falsification

No engineered A/B organelles, reliable universal protein danger classifier, selective alert-delivery mechanism, or safe control of translation/apoptosis have been demonstrated. Molecular function depends on folding, post-translational modification, cell state, and environment. Brownian motion in crowded cytoplasm is not equivalent to independent uniform sampling. The model should be rejected or revised if realistic calibrated parameters and matched resource costs yield no meaningful improvement over natural surveillance, or if false-intervention costs dominate.

Next steps

Implement explicit spatial diffusion and Brownian trajectories, dynamic hysteresis, temporally resolved B signaling, dependent observers, realistic energy/resource accounting, and sensitivity analysis. Seek expert review and biological calibration before considering any experimental program. This model must not be interpreted as a treatment proposal.

Selected background literature

Brandman O & Hegde RS (2016), Ribosome-associated protein quality control, doi:10.1038/nsmb.3147. Karamyshev AL et al. (2014), Cell, doi:10.1016/j.cell.2014.02.053. Turing AM (1952), doi:10.1098/rstb.1952.0012. Elmore S (2007), Apoptosis: A Review of Programmed Cell Death, doi:10.1080/01926230701320337. Snijder B & Pelkmans L (2011), doi:10.1038/nrm3044. These sources supply background, not evidence that the proposed A+B architecture is feasible; bibliographic details should be independently checked before scholarly publication.